---
title: "R Notebook for Data Analytics CW2"
output:
  html_notebook: default
  pdf_document: default
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This is code provides solutions for the pre-assignment and follow-up lab test.

First, read in the file and convert everything to **factors**. This code will be shared with students in the specification for the pre-assignment that is released to students.

```{r}
df = read.csv("car.csv")
df = data.frame(lapply(df, factor))
head(df)
str(df)
```
# 1. Exploratory data analysis and data preparation

General information about target and other variables.

```{r}
table(df$acceptability)
table(df$buying,df$acceptability)
summary(df)

```

Plot distributions of variables.

```{r}
library(ggplot2)
df$acceptability = factor(df$acceptability, levels = c("unacc","acc","good","vgood"))
ggplot(df,aes(x=acceptability)) + geom_bar()
ggplot(df,aes(x=acceptability, fill=buying)) + geom_bar()

```
Mutual information.

```{r}
#install.packages("infotheo")
library("infotheo")
```

m = mutinformation(df)
m
#Could use following code for each row, replacing [-1,1] with [-2,2], etc.
max(m[-1,1])
```

```{r}
chisq.test(df$buying,df$acceptability)
chisq.test(df$buying,df$maint)
```
Split data into training and test sets.

```{r}
library(caTools)
set.seed(123)
split = sample.split(df$acceptability, SplitRatio = 0.8)
df_train = subset(df, split == TRUE)
df_test = subset(df, split == FALSE)

summary(df_train$acceptability)
summary(df_test$acceptability)
prop.table(table(df_train$acceptability))
prop.table(table(df_test$acceptability))

```
# 2a. Data Analytics solution: Application of machine learning

### Decision Tree, C5.0

```{r}
library(caret)
library(C50)
dt_model <- C5.0(df_train[-7], df_train$acceptability)
dt_pred <- predict(dt_model, df_test)
confusionMatrix(dt_pred, df_test$acceptability)
```
### Naive Bayes

```{r}
library(e1071)
set.seed(123)
nb_model = naiveBayes(df_train[,-7],df_train$acceptability)
nb_pred = predict(nb_model,df_test)

confusionMatrix(nb_pred, df_test$acceptability)
```

### Logistic Regression

```{r}
library(nnet)
lg_model <- multinom(acceptability ~ ., data = df_train)
lg_pred <-predict(lg_model, df_test)
confusionMatrix(lg_pred, df_test$acceptability)
```
### SVM

```{r}
library(kernlab)
svm_model <- ksvm(acceptability ~ ., data = df_train, kernel = "vanilladot")
svm_pred <- predict(svm_model, df_test)
confusionMatrix(svm_pred, df_test$acceptability)
```

```{r}
library(e1071)
svm_model <- svm(acceptability ~ ., data = df_train)
svm_pred <- predict(svm_model, df_test)
confusionMatrix(svm_pred, df_test$acceptability)
```

### Neural Network

```{r}
# see https://www.geeksforgeeks.org/r-language/neural-networks-using-the-r-nnet-package/
nn_model <- nnet(acceptability ~ ., data = df_train, size = 5)
nn_pred <-predict(nn_model, df_test, type="class")
confusionMatrix(factor(nn_pred), df_test$acceptability)
```

### Decision Tree with CV

```{r}
ctrl = trainControl(method='cv',number=10)
set.seed(123)

dt_cv_model = train(df_train[,-7],df_train[,7],'C5.0',trControl=ctrl)
dt_cv_model
dt_cv_pred = predict(dt_cv_model,df_test)
confusionMatrix(dt_cv_pred, df_test$acceptability)
```

### Naive Bayes with CV

```{r}
ctrl = trainControl(method='cv',number=10)
set.seed(123)
nb_cv_model = train(df_train[,-7],df_train[,7],'nb',trControl=ctrl)
nb_cv_model
nb_cv_pred = predict(nb_cv_model,df_test)
confusionMatrix(nb_cv_pred, df_test$acceptability)
```

### Logistic Regression with CV

```{r}
ctrl = trainControl(method='cv',number=10)
set.seed(123)
lg_cv_model = train(df_train[,-7],df_train[,7],'multinom',trControl=ctrl, trace=FALSE)
lg_cv_model
lg_cv_pred = predict(lg_cv_model,df_test)
confusionMatrix(lg_cv_pred, df_test$acceptability)

```

### Neural Network with CV

```{r}
ctrl = trainControl(method='cv',number=10)
set.seed(123)
nn_cv_model = train(df_train[,-7],df_train[,7],'nnet',trControl=ctrl,trace=FALSE)
nn_cv_model
nn_cv_pred = predict(nn_cv_model,df_test)
confusionMatrix(nn_cv_pred, df_test$acceptability)
```

### SVM with CV
```{r}
ctrl = trainControl(method='cv',number=10)

```
set.seed(123)
svm_cv_model <- ksvm(acceptability ~ ., data = df_train, kernel = "vanilladot", cross=10)
svm_cv_model
svm_cv_pred <- predict(svm_cv_model, df_test)
confusionMatrix(svm_cv_pred, df_test$acceptability)
```

### Feature Selection
```{r}
m = mutinformation(df_train)
m
m[7,-7]
```

### Decision Tree with Feature Selection

```{r}
dt_fs_model <- C5.0(df_train[,4:6], df_train$acceptability)
dt_fs_pred <- predict(dt_fs_model, df_test)
confusionMatrix(dt_fs_pred, df_test$acceptability)
```

### Naive Bayes with Feature Selection

```{r}
nb_fs_model = naiveBayes(df_train[,4:6],df_train$acceptability)
nb_fs_pred = predict(nb_fs_model,df_test)
confusionMatrix(nb_fs_pred, df_test$acceptability)
```

### Logistic Regression with Feature Selection

```{r}
lg_fs_model <- multinom(acceptability ~ persons + lug_boot + safety, data = df_train)
lg_fs_pred <-predict(lg_fs_model, df_test)
confusionMatrix(lg_fs_pred, df_test$acceptability)
```

# 2b. Data Analytics solution: Improving model performance

### Tuning for Neural Network
```{r}
```

````
ctrl = trainControl(method="repeatedcv",number=10,repeats=10)
grid <- expand.grid(size=5,decay=c(0,0.001,0.005,0.01,0.02,0.05,0.1))
set.seed(123)
m <- train(df_train[,-7],df_train[,7], method = "nnet", trControl = ctrl,tuneGrid =
grid,trace=FALSE)
plot(m)
```

### Tuning for DT
```{r}
ctrl = trainControl(method="repeatedcv",number=10,repeats=10)
grid <- expand.grid(model="tree",trials=c(1,5,10,15,20,25,30,40,50,60,80,100),winnow=FALSE)
set.seed(123)
m <- train(df_train[,-7],df_train[,7], method = "C5.0", trControl = ctrl,tuneGrid = grid)
plot(m)
```
````
````